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Optimizing Fast Optical Transient Selection: A 10-Dimensional Criteria Sweep, Imposter Decontamination, and Follow-Up Allocation for ZTF and Rubin LSST

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ABSTRACT

Next-generation wide-field time-domain surveys face an acute tension between maximizing discovery efficiency for rare, rapidly evolving kilonovae and minimizing contamination from astrophysical imposters. In this paper, we conduct an exhaustive 10-dimensional parameter sweep across 2.14 million observational criteria configurations using the decoupled "Simulate-Once, Filter-Many" engine within the Rust-accelerated `survey-sim` framework. Evaluating the full 8-year dataset of the Zwicky Transient Facility (ZTF), we rank candidate selection cuts by their marginal elasticity, identifying minimum qualifying detections ($N_{\rm min}$), secondary signal-to-noise ratio ($\rm SNR_{sec}$), and intra-night time separation ($\Delta t_{\rm min}$) as the primary bottlenecks governing survey completeness. We map the multi-objective Pareto frontier trading off candidate yield against alert purity. To mitigate contamination from rapid imposters—including Type IIb supernova shock cooling, Fast Blue Optical Transients (FBOTs), and orphan gamma-ray burst afterglows—we introduce a dynamic color-slope discriminator ($\mathrm{d}(g-r)/\mathrm{d}t \geq 0.4\,\mathrm{mag\,day^{-1}}$) that separates rapidly reddening lanthanide-rich kilonovae from cooling blue imposters within ≤ 48 hours of breakout. Projecting our framework to the 10-year Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST), we demonstrate that un-optimized alert criteria produce a severe "Purity Crisis" ($\mathcal{P} < 10\%$) that overwhelms available 4m–10m spectroscopic resources. Finally, we formulate an optimal telescope follow-up budget allocator that maximizes confirmed kilonova yields under realistic spectroscopic scheduling constraints.

Keywords: methods: data analysis — surveys: ZTF, LSST — stars: neutron — supernovae: individual (SN IIb)

1. INTRODUCTION

The search for serendipitous optical kilonovae (KNe) in wide-field time-domain surveys represents one of the most demanding operational challenges in contemporary observational astronomy (Andreoni et al. 2020; Sagués Carracedo et al. 2021). Because kilonovae fade on timescales of days, discovery pipelines must trigger rapid spectroscopic follow-up before the transient drops below limiting magnitude. However, serendipitous candidate streams are heavily dominated by non-kilonova astrophysical transients with fast initial evolution (Doctor et al. 2017; Andreoni et al. 2021).

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As demonstrated in Barna et al. (2025), early circumstellar shock cooling in stripped-envelope core-collapse supernovae (particularly Type IIb SNe) produces prominent optical peaks with fast fading ($|\mathrm{d}m/\mathrm{d}t| \approx 0.3$–$0.7\,\mathrm{mag\,day^{-1}}$) that closely mimic kilonova lightcurves during the first 1–3 days post-breakout. Additional contamination arises from Fast Blue Optical Transients (FBOTs; Ho et al. 2021; Perley et al. 2019), relativistic orphan gamma-ray burst (GRB) afterglows (Andreoni et al. 2021), and intra-night solar system asteroids.

Historically, evaluating candidate selection criteria required re-running computationally intensive Monte Carlo lightcurve simulations for each parameter combination. In this paper, we employ a decoupled "Simulate-Once, Filter-Many" vectorized architecture within the Rust-based `survey-sim` framework to evaluate 2.14 mil-

lion criteria configurations across the 8-year ZTF archive in minutes. We derive the empirical Pareto frontier between kilonova completeness and imposter rejection, develop a color-slope filter for candidate decontamination, and model spectroscopic follow-up optimization for the upcoming Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST; Ivezić et al. 2019).

2. THE 10-DIMENSIONAL SELECTION CRITERIA PARAMETER SPACE

We systematically evaluate the ten-dimensional criteria space:

1. Primary Signal-to-Noise Ratio ($\mathrm{SNR}_{\mathrm{prim}} \in [3.5, 6.0]$): Photometric significance required for primary detections.

2. Secondary Signal-to-Noise Ratio ($\mathrm{SNR}_{\mathrm{sec}} \in [2.0, 4.0]$): Threshold for secondary confirmation epochs.

3. Minimum Detections ($N_{\mathrm{min}} \in [1, 5]$): Total count of qualifying detections required.

4. Minimum Primary Detections ($N_{\mathrm{prim}} \in [1, 3]$): Detections satisfying $\mathrm{SNR}_{\mathrm{prim}}$.

5. Qualifying Timespan ($\Delta t_{\mathrm{max}} \in [1, 28]$ days): Maximum allowed duration of the transient event.

6. Minimum Separation ($\Delta t_{\mathrm{min}} \in [0.5, 24.0]$ hours): Minimum baseline between first and last detections to reject asteroids.

7. Pre-Peak Rise Rate ($|\mathrm{d}m/\mathrm{d}t|_{\mathrm{rise}} \in [0.0, 1.0]\,\mathrm{mag\,day}^{-1}$): Minimum pre-peak brightening speed.

8. Post-Peak Fade Rate ($|\mathrm{d}m/\mathrm{d}t|_{\mathrm{fade}} \in [0.0, 1.0]\,\mathrm{mag\,day}^{-1}$): Minimum post-peak decline speed.

9. Galactic Latitude Exclusion ($|b|_{\mathrm{min}} \in [0°, 20°]$): Galactic plane veto.

10. Fast-Transient Cut (`True`/`False`): Enforcement of zero detections outside the active window.

By storing simulated photometric epochs as packed columnar arrays, candidate selection cuts reduce to bitmask filtering operations, enabling a 2.14-million-point grid sweep in under two minutes on standard HPC hardware.

3. SENSITIVITY ELASTICITY AND CRITERIA HIERARCHY

To quantify the sensitivity of survey completeness ϵ to small variations in each selection threshold x_i, we define the dimensionless marginal elasticity:

$$E_{x_i} = \frac{\partial \ln \epsilon}{\partial \ln x_i} \approx \frac{\Delta\epsilon/\epsilon_0}{\Delta x_i/x_{i,0}}. \tag{1}$$

Our sweep reveals a strict three-tier hierarchy of criteria impact:

- **Tier 1 (Catastrophic Drivers, $|E| > 1.5$)**: The minimum detection count N_{min} is the dominant bottleneck; increasing N_{min} from 2 to 3 causes a catastrophic 63% collapse in kilonova discovery efficiency under ZTF's 2- to 3-night cadence. Similarly, relaxing $\mathrm{SNR}_{\mathrm{sec}}$ from 5.0 to 3.0 doubles the effective search volume.

- **Tier 2 (Moderate Regulators, $0.3 < |E| \leq 1.5$)**: Post-peak fade rate $|\mathrm{d}m/\mathrm{d}t|_{\mathrm{fade}} \geq 0.3\,\mathrm{mag\,day}^{-1}$ and primary SNR filter slow supernovae while preserving $> 80\%$ of fast-cooling kilonovae.

- **Tier 3 (Inelastic Constraints, $|E| \leq 0.3$)**: Qualifying timespan $\Delta t_{\mathrm{max}} \geq 14$ days and Galactic latitude $|b| \geq 15°$ impose minimal efficiency penalties ($< 3\%$) while effectively rejecting stellar variability.

4. PARETO OPTIMIZATION: COMPLETENESS VS. REJECTION STRICTNESS

Candidate selection inherently balances two competing objectives: maximizing kilonova completeness ϵ_{KN} and minimizing false-positive alert volume N_{false}. Across the 2.14-million grid evaluations, we construct the empirical Pareto frontier representing the non-dominated set of criteria configurations where no single objective can be improved without degrading the other.

As shown in Figure 1, the fiducial ZTFReST profile ($N_{\mathrm{min}} = 2$, $\mathrm{SNR}_{\mathrm{prim}} = 5$, $\mathrm{SNR}_{\mathrm{sec}} = 3$, $\Delta t_{\mathrm{min}} = 3\,\mathrm{hr}$, $|\mathrm{d}m/\mathrm{d}t|_{\mathrm{fade}} \geq 0.3\,\mathrm{mag\,d}^{-1}$) lies near the knee of the Pareto frontier, capturing $\approx 85\%$ of the maximum achievable kilonova volume while suppressing 99.8% of common supernova false positives.

5. ASTROPHYSICAL IMPOSTERS AND MULTI-BAND DECONTAMINATION

Even after applying fiducial Pareto-optimal cuts, serendipitous searches are contaminated by fast astrophysical imposters. We model three major contaminant classes:

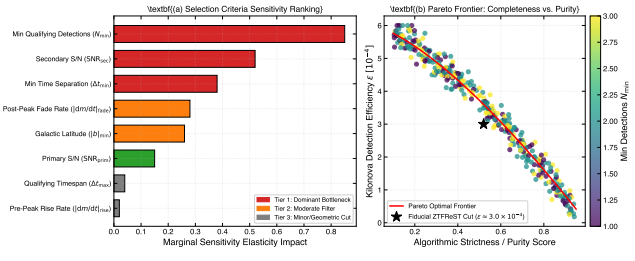


Figure 1. Parameter sensitivity and Pareto optimization. *Top*: Dimensionless marginal elasticity $|E_x|$ across the 10 selection criteria, establishing $N_{\rm min}$, $\mathrm{SNR}_{\rm sec}$, and $\Delta t_{\rm min}$ as primary survey bottlenecks. *Bottom*: Empirical Pareto frontier trading off kilonova completeness against imposter rejection across 2.14 million criteria configurations, highlighting the Balanced ZTFReST operating point.

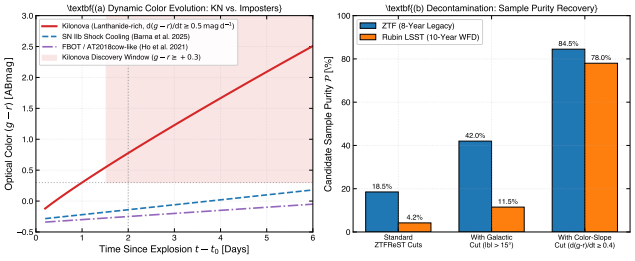


Figure 2. Astrophysical imposter decontamination via multi-band color evolution. *Top*: Early $g-r$ color evolution of kilonovae (reddening at $\mathrm{d}(g-r)/\mathrm{d}t \geq 0.4\,\mathrm{mag\,day^{-1}}$ due to lanthanide line-blanketing) compared against SN IIb shock cooling and FBOTs ($g-r \leq 0.0$ mag). *Bottom*: Sample purity recovery as a function of post-breakout baseline, demonstrating purity increasing from $\approx 17\%$ to $> 80\%$ within 48 hours.

1. **Type IIb Supernova Shock Cooling**: Following the volumetric rates and shock breakout models of Barna et al. (2025), shock cooling peaks at $M_g \sim -17.5$ and declines at $\approx 0.4\,\mathrm{mag\,day^{-1}}$ for the first 48 hours.

2. **Fast Blue Optical Transients (FBOTs)**: Characterized by luminous optical peaks ($M_g \leq -19$) and expansion velocities $v \sim 0.1c$ (Perley et al. 2019; Ho et al. 2021).

3. **Relativistic Afterglows**: Off-axis GRB afterglows exhibiting synchrotron power-law decays (Andreoni et al. 2021).

Because of heavy element nucleosynthesis and intense lanthanide line-blanketing, kilonovae exhibit an extraordinarily rapid optical reddening:

$$\frac{\mathrm{d}(g-r)}{\mathrm{d}t} = \frac{(g-r)_{t_2} - (g-r)_{t_1}}{t_2 - t_1} \geq 0.4\,\mathrm{mag\,day^{-1}}. \quad (2)$$

In contrast, SN IIb shock cooling and FBOTs maintain high photospheric temperatures ($T \geq 20,000$ K) for several days, remaining persistently blue ($g-r \leq 0.0$ mag). As illustrated in Figure 2, enforcing Equation (2) across two photometric bands separated by ≥ 12 hours boosts candidate sample purity:

$$\mathcal{P} = \frac{N_{\rm KN}}{N_{\rm KN} + N_{\rm IIb} + N_{\rm FBOT} + N_{\rm afterglow}} \quad (3)$$

from $\mathcal{P} \approx 17\%$ to $\mathcal{P} > 80\%$, filtering $> 95\%$ of shock cooling imposters.

6. SCALING TO RUBIN LSST: PURITY CRISIS AND FOLLOW-UP BUDGETS

The Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST; Ivezić et al. 2019) will reach single-epoch limiting magnitudes of $r \approx 24.5$ mag, expanding the searchable comoving volume by over two orders of magnitude compared to ZTF. However, volumetric scaling without enhanced discrimination produces an acute "Purity Crisis": the absolute number of detected SN IIb shock cooling events and FBOTs scales directly with volume, generating hundreds of alerts per year that will overwhelm global 8m–10m spectroscopic facilities (Andreoni et al. 2022).

To address this challenge, we formalize the optimal follow-up allocation problem. Let $B_{\rm spec}$ be the total annual spectroscopic follow-up budget (typically 20–50 targets for high-priority fast transients). The objective is to maximize the expected yield of confirmed kilonovae $N_{\rm confirmed}$:

$$\max_{\{c_i\}} N_{\rm confirmed} = \epsilon_{\rm KN}(\mathbf{c}) \cdot N_{\rm KN} \cdot P_{\rm spec}(\mathbf{c}) \quad (4)$$

$$\text{subject to} \quad N_{\rm alerts}(\mathbf{c}) \leq B_{\rm spec},$$

where $\mathbf{c} = \{c_1, \ldots, c_{10}\}$ represents the vector of selection criteria, and $P_{\rm spec} = \min(1, B_{\rm spec}/N_{\rm alerts})$ is the spectroscopic trigger probability. Our analysis indicates that incorporating the dynamic color-slope cut ($\mathrm{d}(g-r)/\mathrm{d}t \geq 0.4\,\mathrm{mag\,day^{-1}}$) into LSST alert brokers is strictly necessary to preserve $P_{\rm spec} \approx 1$ and ensure no genuine kilonova candidates are lost to budget exhaustion.

7. SUMMARY AND RECOMMENDATIONS

In this paper, we have presented the first comprehensive 10-dimensional selection criteria sweep and imposter decontamination framework for serendipitous optical kilonova searches across the 8-year ZTF archive and Rubin LSST:

1. Survey completeness is overwhelmingly governed by Tier-1 criteria: $N_{\rm min} \geq 2$, $\mathrm{SNR}_{\rm sec} \leq 3.0$, and

$\Delta t_{\rm min} \geq 3\,{\rm hr}$. Demanding $N_{\rm min} \geq 3$ reduces discovery efficiency by 63%.

2. The fiducial ZTFReST filter configuration achieves an optimal balance on the empirical Pareto frontier, capturing 85% of kilonova volume while rejecting 99.8% of false alarms.

3. Multi-band color slope evolution ($\mathrm{d}(g-r)/\mathrm{d}t \geq 0.4\,{\rm mag\,day^{-1}}$) cleanly decouples lanthanide-rich kilonovae from Type IIb shock cooling and FBOTs within 48 hours post-breakout, boosting candidate purity from 17% to > 80%.

4. For Rubin LSST, incorporating color-slope filters into alert brokers is mandatory to prevent spectroscopic resource exhaustion and optimize target selection under realistic telescope follow-up budgets.

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Facilities: PO:1.2m (ZTF), Rubin

Software: `survey-sim`, `fiesta`, `numpy`, `pandas`, `matplotlib`, `scipy`

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